CHE210: ***DISCUSSION SESSION 9*** **NAME:**___________________________________

MARCH 9, 2020

Recycle and Purge in the Synthesis of Methanol

Methanol is produced in the reaction of carbon dioxide and hydrogen:

$$CO_2 + 3\ H_2 \rightarrow CH_3OH + H_2O$$

The fresh feed to the process contains hydrogen, carbon dioxide, and 0.400% inerts (I). The reactor effluent passes to a condenser that removes essentially all the methanol and water formed and none of the reactants or inerts. The latter substances are recycled to the reactor. To avoid buildup of the inerts in the system, a purge stream is withdrawn from the recycle.

The feed to the *reactor* (not the fresh feed to the process) contains 28.0 mole% CO_2, 70.0 mol% H_2, and 2.00 mol% inerts. The single-pass conversion of hydrogen is 60.0%. Calculate the molar flow rates and the molar compositions of the fresh feed, the total feed to the reactor, the recycle stream, and the purge stream for a methanol production rate of 155 kmol CH_3OH/hr.